

Acknowledgements

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The BON domain: a putative membrane-binding domain

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A novel conserved protein region – the BON (bacterial OsmY and nodulation) domain – is found in the bacterial osmotic-shock-resistance protein OsmY, a family of haemolysins, a group of nodulation specificity proteins and secretory channels, and several hypothetical proteins. Functional annotation in the literature suggests that it interacts with phospholipid membranes. A lack of catalytic residues in the sequence alignment supports the hypothesis that it is a binding domain.

Scanning genomes for novel domains is an effective method for elucidating both organism-specific information

and more widespread biological processes [1]. We decided to carry out such a scan on *Deinococcus radiodurans* because it is renowned for its ability to withstand massive doses of radiation, and through similar mechanisms it can repair extensive DNA damage caused by rehydration from a desiccated state [2]. Our investigation identified a domain that is involved in osmotic-shock protection and other cell-membrane-localized processes through its interactions with phospholipid membranes.

Identification of the conserved region

A conserved repeat was identified in *D. radiodurans* protein DR0888 (SWISS-PROT accession number: Q9RVY3) using Prospero (<http://www.well.ox.ac.uk/rmott/ARIADNE/prospero.shtml>) and subsequent to

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masking low complexity regions using 'seg' [3]. Residues 9–75 aligned to 124–190 with an E-value of 1.2×10^{-6} . The aligned pair was then searched against SWISS-PROT and TrEMBL [4] using HMMER (<http://hmmer.wustl.edu/>). The initial search found a suggestive (E-value = 0.16) match to *Xanthomonas axonopodis* protein XAC0682 (Q8PPK4), residues 53–116. This region was then used to initiate a set of iterative HMMER searches. Both global (ls) and fragment (fs) models were built and searched with, and results combined using E-value cut-offs of 0.1 and 0.01, respectively. Alignments were visually examined and potential false-positives removed between rounds. T-Coffee [5] and manual editing were used to produce the final alignment. After 13 rounds the searches converged to identify 61 proteins, including both the regions from DR0888, confirming the validity of the initial suggestive match. To ratify the searches, an equivalent process was carried out at the National Center for Biotechnology Information (NCBI) PSI-BLAST server (<http://www.ncbi.nlm.nih.gov/BLAST/>) using an E-value cut-off of 0.02. The occurrence of these regions as singlets in some proteins implies that they are structurally stable in isolation and are true domains. The domain was termed the BON (bacterial OsmY and nodulation) domain.

OsmY comprises two BON domains

The BON domain is typically ~60 residues long and has an α/β predicted fold (Fig. 1). There is a conserved glycine residue and several hydrophobic regions. This pattern of

conservation is more suggestive of a binding or structural function rather than a catalytic function.

The OsmY protein is an *Escherichia coli* 20 kDa outer membrane or periplasmic protein [6] that has RpoS-controlled expression in the stationary phase under normal growth conditions [7]. It is also expressed in response to a variety of stress conditions, in particular, helping to provide protection against osmotic shock [6,8,9]. One hypothesis is that OsmY prevents shrinkage of the cytoplasmic compartment by contacting the phospholipid interfaces surrounding the periplasmic space [9,10]. This would physically prevent the inner membrane from shrinking by attaching it to the more rigid outer membrane. The domain architecture of two BON domains alone suggests that these domains contact the surfaces of phospholipids, with each domain contacting a membrane.

Notably, a group of putative haemolysins also consist of two BON domains. The assignment of haemolytic activity is based on the conferment of haemolytic activity to *E. coli* after transformation with a plasmid that contained DNA sequence from *Actinobacillus pleuropneumoniae* [11]. To our knowledge, no other work has been carried out to confirm that the encoded protein (HLY) is directly responsible for this activity; however, the ability to interact with cell membranes would be expected of a haemolysin. Although both OsmY and HLY have identical domain architectures, the two subfamilies can be distinguished by phylogenetic analysis (data not shown).

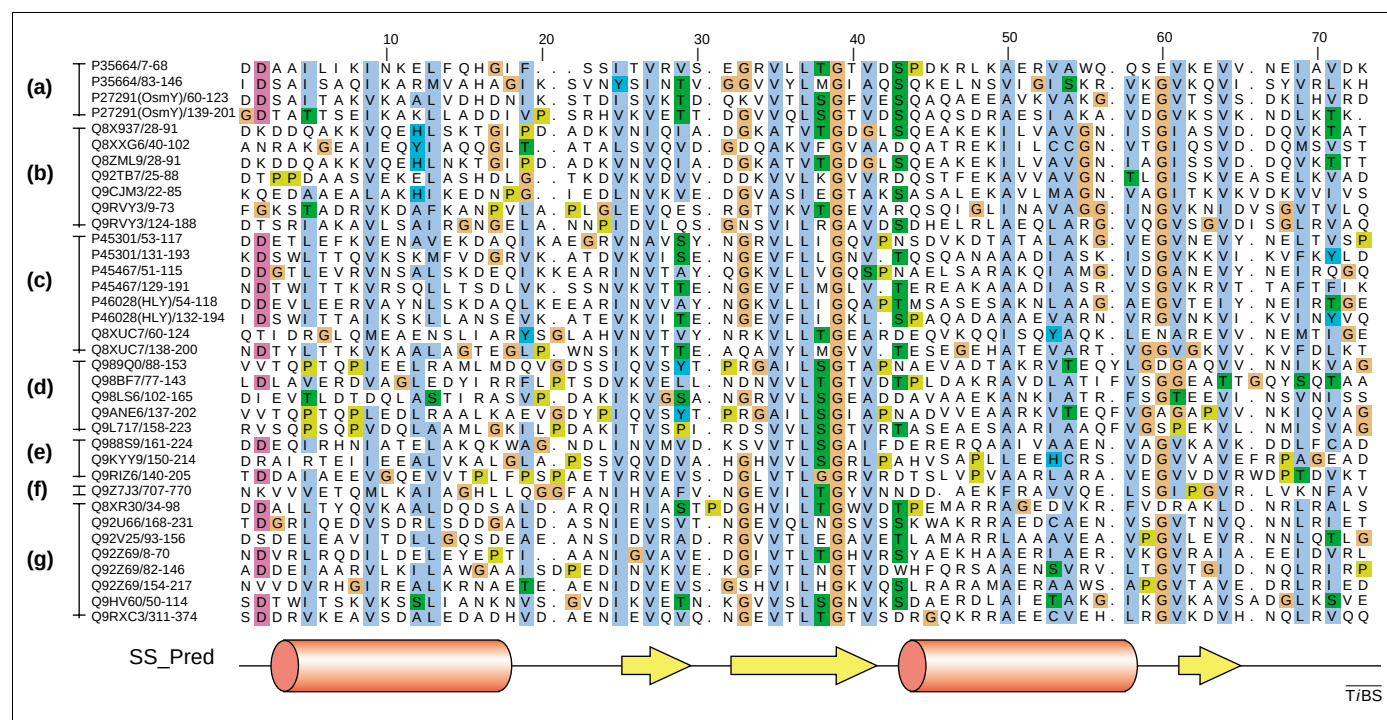


Fig. 1. Representative alignment of BON (bacterial OsmY and nodulation) domains aligned with T-Coffee and displayed using the Jalview viewer (M. Clamp, unpublished; www2.ebi.ac.uk/~michele/jalview) and ClustalX colouring [18]. Sequence names are SWISS-PROT identifiers. A predicted secondary structure based on results from the Jpred server (<http://www.compbio.dundee.ac.uk/~www-jpred>) is shown below the alignment: yellow arrows indicate β strands, red cylinders indicate α helices. The sequences are clustered into seven groups based on function and domain architecture: (a) OsmY protein-related (Swiss: P27291); (b) LysM domain-associated (Pfam: PF01476); (c) HLY protein-related (Swiss: P46028); (d) nodulation specificity; (e) CBS domain-associated (Pfam: PF00571); (f) regulatory; and (g) other. This multiple sequence alignment (alignment number ALIGN_000546) has been deposited with the European Bioinformatics Institute (ftp://ftp.ebi.ac.uk/pub/databases/embl/align/ALIGN_000546).

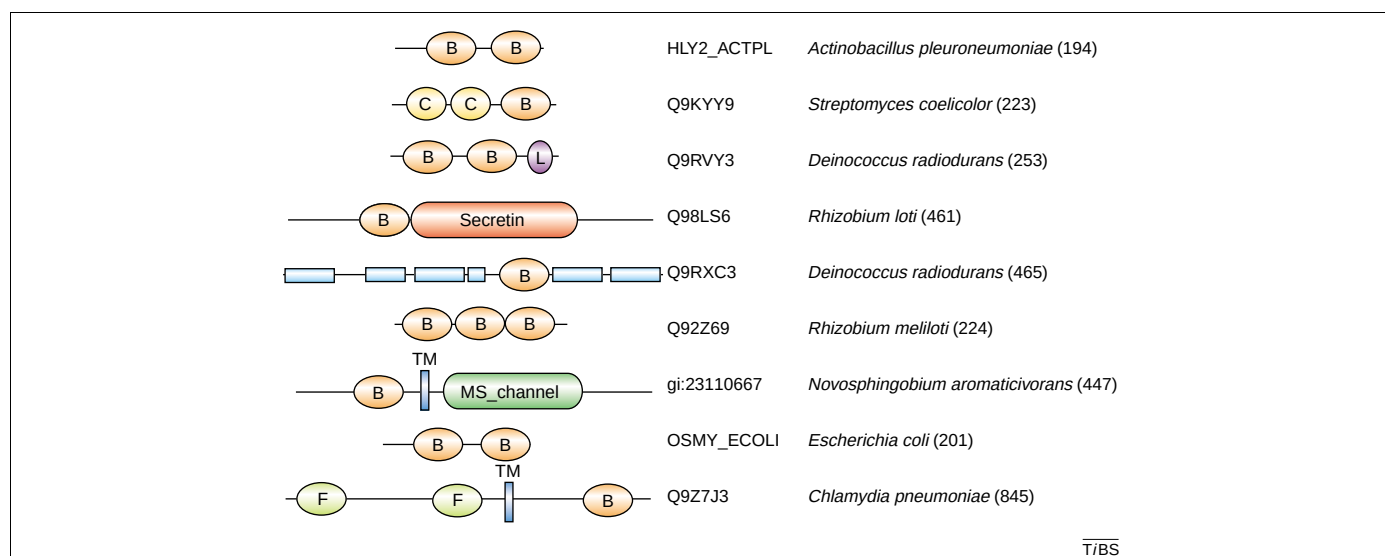


Fig. 2. Example domain architectures are shown in the 'beads on string' view. Proteins are identified with SWISS-PROT accessions and identifiers except gi:23110667, which is a GenBank identifier [19]. Numbers in brackets are the number of residues. Domains were identified using Pfam [20]; accession numbers are shown in brackets after the domain name. Domains shown: BON (B; PF04972), orange; Secretin (PF00263), red; LysM (L; PF01476), purple; CBS (C; PF00571), yellow; MS_channel (PF00924), green; FHA (F; PF00498), light green. The light blue boxes shown for Q9RXC3 are low-complexity regions as determined using 'seg' [4]. The TM regions (dark blue) are transmembrane regions predicted using TMHMM [21]. Notably, OsmY has the same architecture as Hly2, but very different biological functions. Abbreviations: BON, bacterial OsmY and nodulation; FHA, forkhead-associated; MS_channel, mechanosensitive channel.

Other BON-containing proteins

The other occurrences of BON further support the hypothesis of it associating with phospholipid membranes (Fig. 2). It occurs in association with two membrane-pore-forming domains - Secretin and MS_channel [12–14]. The specific Secretin family includes the NolX protein; NolX is involved in the secretion of the pilus that determines host specificity for rhizobial nodule formation. MS_channel proteins are mechanosensitive ion channels and have been implicated in osmotic regulation; some appear to function in response to membrane deformation [15]. None of the BON-containing MS_channel proteins have been specifically characterized; however, it is possible that the BON domain reacts to deformations in the plasma membrane, and allosterically signals to the ion channel domain. It is also co-occurs with the cell-wall peptidoglycan-associating LysM domain [16]. Finally, it is found in a set of *Chlamydia* putative regulatory proteins (e.g. Q9Z7J3). These proteins have homology to an adenylate cyclase, but this is because they also contain a forkhead-associated (FHA) domain [17] (data not shown). Nothing is known about the function of this family of regulators.

Phyletic distribution

Most proteobacteria seem to possess one or two BON-containing proteins, typically of the OsmY-type proteins (data not shown); outside of this group the distribution is more disparate. Also, the family is unusually expanded in *Burkholderia*, a genus containing several significant mammalian pathogens with varying host ranges. The number of BON-domain proteins varies between *Burkholderia* species, suggesting that they perform specific roles in their respective lifestyles. *B. pseudomallei* has eight BON-containing proteins (M. Holden, pers. commun.). Included in this set are a NolX homologue and two proteins with an

unusual three tandemly repeated BON domains. Homologues of this protein are only found in some other *Burkholderia* and in the main symbiosis or pathogenicity plasmids of *Ralstonia* (plant pathogens) and *Rhizobiales*. This distribution suggests that these proteins play a role in host invasion or host-cell interactions. Within the completed genomes, no clear operon structures associated with BON domains were found, except in the case of the Rhizobial pilus secretion operon. In conclusion, the BON domain is likely to be a phospholipid-binding domain that is involved in a variety of biological processes.

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Letters

Response: The mitochondrial DNA replication bubble has not burst[☆]

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We cannot address all the issues raised in the opinion piece by Bogenhagen and Clayton [1] in the space available here. Therefore, we have confined our remarks to a few key issues. Bogenhagen and Clayton apparently fail to grasp the central point of our recent *Cell* paper [2], namely that the majority of mitochondrial DNA (mtDNA) replication intermediates are fully double stranded, even though they contain extensive ribonucleotide patches that prevent restriction digestion. These give rise to slow-moving replication-fork arcs that are neither destroyed by single-strand-specific nucleases nor that associate with single-strand-binding proteins. Thus, mtDNA replication intermediates are essentially double stranded. In intact mtDNA preparations, replication intermediates with extensive single-stranded character are virtually absent, but those composed only of double-stranded DNA are also a minority. The absence of extensively single-stranded forms cannot be explained by branch migration because this would yield a prominent arc of heterogeneous single-stranded DNA molecules. Such forms, with the exception of 7S DNA (Supplementary information of Ref. [2]), are simply not seen unless the material is first digested by degradative enzymes.

Furthermore, the slow-moving arcs that we observed cannot be simple transcription intermediates because these would contain extended single-stranded regions, whether of RNA or displaced DNA (with the nascent transcript bound in RNA–DNA hybrid). However, they might arise from transcription of mtDNA molecules that

are also undergoing replication. RNA–DNA hybrids of this type would create priming sites for lagging-strand initiation, the distribution of which would depend on the relative rates of transcription, RNA processing or turnover, and replication fork progression in different portions of the genome.

Bogenhagen and Clayton have failed to notice that our most recent data do not, in fact, refute all aspects of the traditional model: rather they refine it, and reconcile it with the indisputable presence of a minority of fully double-stranded replication intermediates composed only of DNA. The intermediates comprising pure DNA would logically include molecules that had initially contained extensive RNA tracts but which are subsequently removed during maturation. Maturation should lead to fully replicated mtDNA molecules which, we fully agree, retain only a very small number of (non-randomly distributed) ribonucleotides. Note that none of the references cited by Bogenhagen and Clayton (Refs 14–18 in their article) concern the analysis of mtDNA replication intermediates.

Two other features of the traditional model are in no way contradicted by our data. First, the existence of the D loop as a prominent non-linear form of mtDNA is indisputable. However, neither we nor any previous author has demonstrated a clear precursor–product relationship between the D-loop form and productive mtDNA replication. The D-loop form remains an enigma. Second, the existence of an L-strand intermediate of 10.5 kb running from the sites formerly designated as origins of replication O_H and O_L (Ref. 9 in their article) is completely consistent with our inference that both sites function as replication barriers. Molecules initiated by random RNA-priming on the L strand should mature into

[☆] In response to the Opinion article by Daniel Bogenhagen and David Clayton (also in this issue).

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